Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Rejection Rule	$\|z_0\| \geq z_{\alpha/2}$	$z_0 > z_\alpha$	$z_0 < -z_\alpha$

Example 1.9

A motor vehicle manufacturing company desires to introduce a new model motor vehicle. The company claims that the mean fuel consumption of its new model vehicle is lower than that of the existing model of the motor vehicle, which is 27 kms/litre. A sample of 100 vehicles of the new model vehicle is selected randomly and their fuel consumptions are observed. It is found that the mean fuel consumption of the 100 new model motor vehicles is 30 kms/litre with a standard deviation of 3 kms/litre. Test the claim of the company at 5% level of significance.

Solution:

Step 1 : Let the fuel consumption of the new model motor vehicle be assumed to be distributed according to a distribution with mean and standard deviation respectively μ and σ. The null and alternative hypotheses are

Null hypothesis H_0: $\mu = 27$

i.e., the average fuel consumption of the company's new model motor vehicle is not significantly different from that of the existing model.

Alternative hypothesis H_1: $\mu > 27$

i.e., the average fuel consumption of the company's new model motor vehicle is significantly lower than that of the existing model. In other words, the number of kms by the new model motor vehicle is significantly more than that of the existing model motor vehicle.

Step 2 : **Data:**

The given sample information are

Size of the sample (n) = 100. Hence, it is a large sample.

Sample mean ($\bar{x}$)= 30

Sample standard deviation(s) = 3

Step 3 : **Level of significance**

$\alpha = 5\%$

Step 4 : **Test statistic**

The test statistic under H_0 is

$$Z = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}.$$

Since n is large, the sampling distribution of Z under H_0 is the $N(0,1)$ distribution.

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